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PMC9059666 | 羰基化 | 采集状态 imported_unchecked

标题	A new bifunctional heterogeneous nanocatalyst for one-pot reduction-Schiff base condensation and reduction – carbonylation of nitroarenes
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出处	RSC Advances 2019
DOI	10.1039/c8ra10212k
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC9059666/
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摘要（原文节选）

In this work, synthesis of Pd – NHC- γ -Fe2O3-n-butyl-SO3H and its activity as a bifunctional heterogeneous nanocatalyst containing Pd – NHC and acidic functional groups, are described. This newly synthesized nanomagnetic catalyst is fully characterized by different methods such as FT-IR, XPS, TEM, VSM, ICP and TG analysis. At first, the catalytic activity of Pd – NHC- γ -Fe2O3-n-butyl-SO3H is evaluated for the reduction of nitroarenes in aqueous media using NaBH4 as a clean source of hydrogen generation at ambient temperatu ⋯

数据表 · 表 1（12 行，含数值行 12 行）

Reduction of nitroarenes by NaBH4 catalyzed by Pd – NHC- γ -Fe2O3-n-butyl-SO3H in aqueous media at ambient temperature

Entrya	Nitroarene	Time (min)	Isolated yield (%)	Obtained mp (° C)	Reported mp (° C) [ref.]
1	Nitrobenzene	30	98	—b	—
2	4-Nitroaniline	30	97	142 – 143	138 – 142 [23]
3	4-Nitrophenol	30	95	187 – 189	186 – 187 [24]
4	3-Nitrophenol	30	92	123 – 124	120 – 121 [24]
5	2-Nitrophenol	60	92	172 – 173	171 – 173 [25]
6	1-Methyl-4-nitrobenzene	30	70	42 – 43	43 – 44 [23]
7	1-Chloro-4-nitrobenzene	20	98	65 – 67	67 – 70 [23]
8	1-Bromo-4-nitrobenzene	20	97	63 – 64	60 – 62 [23]
9	1-Methoxy-4-nitrobenzene	20	99	55 – 56	57 [24]
10	4-Nitrobenzenesulfonamide	30	98	166 – 167	166 [24]
11	(2-Nitrophenyl)methanol	25	99	80	81 – 83 [23]
12	5-Nitro-1H-indole	35	95	132 – 134	130 [24]

脚注（逐字）：aReaction conditions: NaBH4 (3 equiv.), Pd – NHC- γ -Fe2O3-n-butyl-SO3H (0.6 mol%).bLiquid.

正文对应句（逐字）：In order to investigate the catalytic activity of Pd – NHC- γ -Fe2O3-n-butyl-SO3H as a bifunctional heterogeneous catalyst for the one-pot reduction-Schiff base condensation and one-pot reduction – carbonylation of various nitroarenes, we have initially evaluated the efficiency of the catalyst for the red

数据表 · 表 2（13 行，含数值行 13 行）

One-pot reduction-Schiff base condensation of nitroarenes with arylaldehydes catalyzed by Pd – NHC- γ -Fe2O3-n-butyl-SO3H as a bifunctional heterogeneous catalysta

Entry	Aldehyde	Nitroarene	Time (min)	Isolated yield (%)	Obtained mp (° ° C)	Reported mp (° C) [ref.]
1	Benzaldehyde	3-Nitrophenol	100	94	151	149 [26]
2	Benzaldehyde	3-Nitrophenol	110	98	192 – 194	197 – 198 [27]
3	4-Chlorobenzene	1-Methoxy-4-nitrobenzene	110	87	124 – 125	121 – 123 [28]
4	Benzaldehyde	1-Bromo-4-nitrobenzene	110	74	61 – 62	62 – 63 [27]
5	Benzaldehyde	1-Chloro-4-nitrobenzene	110	71	58 – 60	58 – 61 [29]
6	2-Hydroxybenzaldehyde	Nitrobenzene	130	92	51 – 52	52 – 54 [30]
7	Benzaldehyde	Nitrobenzene	70	93	47 – 49	48 – 50 [31]
8	4-Chlorobenzaldehyde	Nitrobenzene	100	84	59 – 61	60 – 61 [31]
9	4-Cyanobenzaldehyde	Nitrobenzene	90	86	94 – 95	97 – 98 [32]
10	4-Bromobenzaldehyde	Nitrobenzene	100	75	75 – 77	71 – 74 [29]
11	2,4-Dichlorobenzaldehyde	Nitrobenzene	180	87	74 – 76	78 – 80 [31]
12	Nicotinaldehyde	Nitrobenzene	180	90	19 – 20	21.5 [33]

Entry	Aldehyde	Nitroarene	Time (min)	Isolated yield (%)	Obtained mp (°C)	Reported mp (°C) [ref.]
13	1H-Indole-3-carboxaldehyde	Nitrobenzene	240	60	128 – 130	132 [34]

脚注（逐字）：aNitroarene, NaBH4 (3 equiv.) and catalyst was stirred in aqueous media at room temperature for 30 min. Aldehyde (1.2 equiv.) was added to the stirring mixture.

正文对应句（逐字）：In order to investigate the catalytic activity of Pd – NHC- γ -Fe2O3-n-butyl-SO3H as a bifunctional heterogeneous catalyst for the one-pot reduction-Schiff base condensation and one-pot reduction – carbonylation of various nitroarenes, we have initially evaluated the efficiency of the catalyst for the red

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